# Mathematical Models for Genreguessr and Genrerace

Cartography & BGMs

Working note

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**Abstract**

Both games start from the same finite collection of musical genres. A chosen geometry determines neighborhoods: Genreguessr hides a genre and asks the player to identify it through a cluster, while Genrerace asks the player to navigate between neighborhoods. This note specifies the three geometries, proves their metric properties with the necessary qualifications, derives the exact cluster-sampling law, and formalizes the single-player and multiplayer rules. It describes the implemented model rather than proposing a new audio-similarity model.

## 1 The common space and the source data

Let $V = \{0, \dots, N-1\}$ be the set of genre identifiers, with $N = 6291$ in the current Every Noise snapshot. Each $i \in V$ has

$$p_i = (x_i, y_i) \in \mathbb{R}^2, \qquad c_i = (r_i, g_i, b_i) \in \{0, \dots, 255\}^3,$$

together with an ordered list of musical samples. The coordinates and RGB values come from the same cached main map as the graph's genre universe.

The light neighborhood box on each source genre page specifies directed *near* links; the dark box specifies directed *opposite* links. The engine removes self-links and symmetrizes each layer separately:

$$\{i, j\} \in E_\sigma \quad \Longleftrightarrow \quad i \neq j \text{ and } (i \to j \text{ or } j \to i) \text{ is recorded in layer } \sigma.$$

Here $\sigma \in \{n, o\}$. Thus the model is a two-layer graph $(V, E_n, E_o)$. An edge may occur in both layers; those two contributions are retained.

| Quantity in the current local snapshot | Value |
|---|---|
| Genres; distinct combined XYRGB vectors | 6291; 6291 |
| Undirected near edges; opposite edges | 83881; 114896 |
| Edges shared by both layers | 32 |
| Near-graph component sizes | 6280, $\underbrace{1, \dots, 1}_{\text{11 isolates}}$ |
| Components of the union graph $(V, E_n \cup E_o)$ | 1 |

These are observations from the validated local catalog, checked for this note, not assumptions about every future Every Noise release.

All normalizations, matrices and neighborhood searches use the *entire* set $V$. The names' random screen positions are only a display convention. They are not $p_i$, a kernel embedding, or an additional metric.

# 2 Three geometries

## 2.1 XYRGB: a weighted Euclidean geometry

For a block $z_1, \dots, z_N \in \mathbb{R}^r$, define the distinct-pair squared scale

$$s_z^2 = \frac{1}{\binom{N}{2}} \sum_{i<j} \|z_i - z_j\|^2 = \frac{2}{N-1} \sum_{i=0}^{N-1} \|z_i - \bar{z}\|^2, \qquad \bar{z} = \frac{1}{N} \sum_i z_i. \tag{1}$$

The equality follows by expanding squared distances: $\sum_{i<j} \|z_i - z_j\|^2 = N \sum_i \|z_i - \bar{z}\|^2$. It permits computation of the scale without storing all pairs. For a constant block the implementation uses the effective scale $s_z^2 = 1$; that block still contributes zero.

Using $s_p, s_c$ for the position and color scales, respectively, set

$$d_\alpha(i,j)^2 = \alpha \frac{\|p_i - p_j\|^2}{s_p^2} + (1-\alpha) \frac{\|c_i - c_j\|^2}{s_c^2}, \qquad 0 < \alpha < 1. \tag{2}$$

Both spatial axes share one scale, preserving the aspect ratio of the source geometry. Normalizing $x$ and $y$ separately would define a different model. The default game setting is $\alpha = 0.8$.

**Proposition 2.1.** *The function $d_\alpha$ is a Euclidean pseudometric on genre identifiers. It is a metric precisely when the combined vectors $(p_i, c_i)$ are distinct.*

*Proof.* Define

$$\Phi_\alpha(i) = \left( \frac{\sqrt{\alpha}}{s_p} p_i, \ \frac{\sqrt{1-\alpha}}{s_c} c_i \right) \in \mathbb{R}^5.$$

Then $d_\alpha(i,j) = \|\Phi_\alpha(i) - \Phi_\alpha(j)\|$. Euclidean distance supplies symmetry and the triangle inequality; positivity of both weights gives the stated identity-of-indiscernibles condition. □

Consequently, the current snapshot gives a genuine metric for every $0 < \alpha < 1$. For a future snapshot with duplicate vectors it would be a pseudometric, or a metric on the quotient identifying those duplicates.

When both blocks are nonconstant, uniformly sampling a distinct pair gives

$$\mathbb{E}_{i \neq j}[d_\alpha(i,j)^2] = 1.$$

The mean squared contributions of position and color are $\alpha$ and $1 - \alpha$, respectively. This is an *average squared* balance: $\alpha > 1/2$ does not make position dominate for every pair.

**Example 2.2.** If a pair has normalized position distance 0.5 and normalized color distance 1, then at $\alpha = 0.8$,

$$d_\alpha = \sqrt{0.8(0.5)^2 + 0.2(1)^2} = \sqrt{0.4}.$$

A pair with the same position but a large color difference can still be far apart. The weight does not discard either block.

## 2.2 The near graph: shortest paths

Write $G_n = (V, E_n)$, with every edge having length one. Define

$$d_G(i,j) = \begin{cases} \text{minimum number of edges in a path from } i \text{ to } j, & \text{if a path exists,} \\ +\infty, & \text{otherwise.} \end{cases} \tag{3}$$

The zero-length path gives $d_G(i,i) = 0$. Concatenating paths proves the triangle inequality; reversing paths proves symmetry. Thus $d_G$ is a finite metric on each connected component and an *extended metric* on all of $V$.

Its canonical neighborhood is the closed unit ball

$$B_1(i) = \{j \in V : d_G(i,j) \le 1\} = \{i\} \cup \{j : \{i,j\} \in E_n\}.$$

The ball's size is $1 + \deg_{G_n}(i)$, not a fixed number of genres. An isolated genre remains in the space and has a singleton ball.

This model discards map positions, RGB values and the opposite layer. A short path expresses a chain of recorded neighborhood relationships; it does not assert that audio signals are close in a separately defined acoustic distance.

### 2.3 The signed kernel: global near/opposite geometry

Let $A_n, A_o \in \mathbb{R}^{N\times N}$ be the symmetric zero-diagonal adjacency matrices of the two layers, and let

$$D_n = \operatorname{diag}(A_n\mathbf{1}), \qquad D_o = \operatorname{diag}(A_o\mathbf{1}).$$

For opposition weight $\lambda > 0$ and regularization $\eta > 0$, define

$$L_s = D_n - A_n + \lambda(D_o + A_o), \qquad K = (L_s + \eta I)^{-1}. \tag{4}$$

The game defaults are $\lambda = 1, \eta = 0.1$. The plus sign in $D_o + A_o$ is essential: opposite edges penalize equal rather than opposite signs.

**Proposition 2.3** (Signed energy)**.** *For $f \in \mathbb{R}^N$, summing each unordered edge once,*

$$f^\top L_s f = \sum_{\{i,j\}\in E_n} (f_i - f_j)^2 + \lambda \sum_{\{i,j\}\in E_o} (f_i + f_j)^2. \tag{5}$$

*Hence $L_s \succeq 0$, $L_s + \eta I \succ 0$, and $K \succ 0$.*

*Proof.* Expand each square. The squared terms produce the degree diagonals; the cross terms produce $-A_n + \lambda A_o$ in the quadratic form. Every summand is nonnegative, and adding $\eta \|f\|^2$ makes the energy strictly positive for $f \neq 0$. □

For intuition, the response to a signal $b \in \mathbb{R}^N$ is

$$Kb = \arg\min_{f\in\mathbb{R}^N} \left\{ \frac{1}{2} f^\top L_s f + \frac{\eta}{2}\|f\|^2 - b^\top f \right\}.$$

Near links favor agreement, opposite links favor sign reversal, and regularization discourages large responses. Inverting the full operator propagates these preferences through the whole graph.

Normalize the kernel and use the resulting chord distance:

$$\rho_{ij} = \frac{K_{ij}}{\sqrt{K_{ii}K_{jj}}}, \qquad d_K(i,j) = \sqrt{2 - 2\rho_{ij}}. \tag{6}$$

This is not the unnormalized kernel distance $\sqrt{K_{ii} + K_{jj} - 2K_{ij}}$; diagonal normalization changes the geometry.

**Proposition 2.4** (A genuine metric on the genre IDs)**.** *In exact arithmetic, $d_K$ is a metric. Moreover,*

$$d_K(i,i) = 0, \qquad 0 < d_K(i,j) < 2 \quad (i \neq j).$$

*Proof.* With $e_i$ the $i$-th standard basis vector, put

$$u_i = \frac{K^{1/2} e_i}{\sqrt{K_{ii}}}.$$

Then $\|u_i\| = 1$, $\langle u_i, u_j \rangle = \rho_{ij}$, and $d_K(i,j) = \|u_i - u_j\|$. Thus symmetry and the triangle inequality are Euclidean. For $i \neq j$, positive definiteness of the corresponding $2 \times 2$ principal submatrix gives $K_{ij}^2 < K_{ii} K_{jj}$, so $-1 < \rho_{ij} < 1$. □

**Example 2.5** (Two genres, exactly)**.** Let $a \in \{0,1\}$ indicate a near edge and $b \in \{0,1\}$ an opposite edge between two vertices. Direct inversion gives

$$\rho_{12} = \frac{a - \lambda b}{\eta + a + \lambda b}, \qquad d_K(1,2)^2 = \frac{2(\eta + 2\lambda b)}{\eta + a + \lambda b}.$$

A near edge alone gives $d_K^2 = 2\eta/(\eta+1) < 2$; an opposite edge alone gives $d_K^2 = 2(\eta+2\lambda)/(\eta+\lambda) > 2$. If both edges occur and $\lambda = 1$, the distance is exactly $\sqrt{2}$. Indeed the overlapping pair contributes

$$(f_1 - f_2)^2 + (f_1 + f_2)^2 = 2f_1^2 + 2f_2^2.$$

The cross terms cancel but the diagonal energy remains. Deleting the overlapping pair would lose this energy and, in a larger graph, change its relations to other vertices.

**What the parameters do.** If $L_s = Q \operatorname{diag}(\nu_1, \ldots, \nu_N) Q^\top$, then

$$K = Q \operatorname{diag}\Big((\nu_1 + \eta)^{-1}, \ldots, (\nu_N + \eta)^{-1}\Big) Q^\top.$$

Low-energy directions receive greater weight. For fixed $\lambda$, as $\eta \to \infty$, normalized off-diagonal correlations tend to zero and all distinct-genre distances tend to $\sqrt{2}$. Very large regularization therefore removes most relational contrast. For finite positive $\eta$, an opposite edge does not prescribe a fixed distance; a pair's distance can depend on relationships throughout the graph.

An overlap forces both $f_i = f_j$ and $f_i = -f_j$ if the energy is zero, so both values must vanish. Zero then propagates along the union graph. Since the current union graph is connected and contains overlaps, $L_s$ is already positive definite for every $\lambda > 0$. Regularization still changes the geometry and improves the spectral condition number:

$$\kappa_2(L_s + \eta I) = \frac{\nu_{\max} + \eta}{\nu_{\min} + \eta}.$$

For other datasets, $L_s$ can be singular; $\eta > 0$ guarantees invertibility in either case.

If the union graph has disconnected components, $K$ is block diagonal, and cross-component distances equal $\sqrt{2}$, not $+\infty$. Conversely, a distance of $\sqrt{2}$ merely means $K_{ij} = 0$; cancellation can also produce this within a connected component.

## 3 From distances to neighborhoods

Fix the metric parameters once for a game. For XYRGB and signed kernel, let $k \in \{5, \ldots, 25\}$. The cluster at $a$ is

$$C_k(a) = \{a\} \cup \{\text{the first } k-1 \text{ elements of } V \setminus \{a\}, \text{ ordered by } (d(a,v), v)\}. \tag{7}$$

Stable genre IDs break ties. The center is included explicitly, even if a future XYRGB snapshot has duplicate feature vectors. For graph mode, set $C(a) = B_1(a)$.

A nearest-$k$ cluster is only *ball-like*: its radius adapts to local density, and a tie at the boundary can leave out equally distant vertices. It need not be a closed metric ball. Neither nearest-$k$ clusters nor unit balls form a partition; they can overlap.

### 3.1 Separating candidate centers

Genreguessr offers $n \in \{3, \dots, 10\}$ clusters. In a continuous mode, centers $a, b$ are compatible when $d(a, b) \geq \ell$, where $\ell \geq 0$ is the chosen separation threshold.

Graph mode instead compares the *sets* $B_1(a), B_1(b)$. For nonempty finite sets $A, B \subseteq V$, define the extended Hausdorff distance

$$H_G(A, B) = \max\left\{\max_{x \in A} \min_{y \in B} d_G(x, y),\ \max_{y \in B} \min_{x \in A} d_G(x, y)\right\}. \tag{8}$$

Compatibility means $H_G(B_1(a), B_1(b)) \geq m$ for an integer threshold $m \geq 0$. This is not the minimum gap between the two sets.

**Proposition 3.1** (A useful unit-ball bound)**.** *If $q = d_G(a, b) < \infty$, then*

$$\max\{q - 1, 0\} \leq H_G(B_1(a), B_1(b)) \leq q.$$

*For $q \geq 1$ the Hausdorff distance is either $q - 1$ or $q$. If $q = +\infty$, it is $+\infty$.*

*Proof.* For a vertex $x$ in the component of $b$,

$$d_G(x, B_1(b)) = \max\{d_G(x, b) - 1, 0\}.$$

The triangle inequality gives the lower bound; the vertex one step from $b$ toward $x$ along a shortest path gives equality. Taking $x = a$ gives the lower Hausdorff bound. For every $x \in B_1(a)$, $d_G(x, b) \leq q + 1$, so its distance to $B_1(b)$ is at most $q$. Interchange $a, b$ for the other direction. Integrality gives the two-value conclusion. Balls at disconnected centers stay in disjoint components. □

The pullback $a, b \mapsto H_G(B_1(a), B_1(b))$ is an extended *pseudometric* on centers: two distinct vertices can have the same unit ball. For example, the two vertices of a single-edge graph have identical balls and Hausdorff distance zero.

On the path $0 - 1 - 2 - 3 - 4$, the balls at 1 and 3 overlap at 2 but have Hausdorff distance 2. Thus even separation $m = 2$ permits overlap. In general, unit balls overlap exactly when their centers' graph distance is at most 2.

## 4 Genreguessr: sampling conditional on a hidden target

The target $T$ is first drawn uniformly from $V$:

$$\mathbb{P}(T = t) = \frac{1}{N}.$$

The draw occurs before checking audio availability or the feasibility of a cluster arrangement. Failure does not silently replace the target.

Fix a target $t$, all parameters, and the induced clusters $C(a)$. Let $a \sim b$ mean that distinct centers meet the selected separation rule. Define

$$A_t = \{a \in V : t \in C(a)\}.$$

This is a *reverse membership* set. The correct cluster's center need not equal $t$. In nearest-neighbor geometry, $A_t$ need not equal $C(t)$.

The valid center sets are

$$\mathcal{F}_t = \{S \subseteq V : |S| = n, \quad |S \cap A_t| = 1, \quad a \sim b \text{ for every distinct } a, b \in S\}. \tag{9}$$

Exactly one offered cluster must contain the target; other overlaps are allowed. The required distribution is uniform on $\mathcal{F}_t$, conditional on finding an arrangement.

For each possible correct center $a \in A_t$, set

$$B_a = \{b \in V \setminus A_t : a \sim b\}, \qquad w_a = \binom{|B_a|}{n-1}, \qquad W_t = \sum_{a \in A_t} w_a,$$

with the binomial coefficient zero when $|B_a| < n - 1$. If $W_t = 0$, no valid set exists. Otherwise, one proposal is:

1. Choose $a \in A_t$ with probability $w_a/W_t$.
2. Choose a uniform unordered subset $D \subseteq B_a$ of size $n - 1$.
3. Accept $S = \{a\} \cup D$ if every pair in $D$ is compatible. Otherwise discard the *entire* proposal and start again.

The engine uses integer combinatorial weights rather than floating-point approximations.

**Proposition 4.1** (Exact conditional uniformity)**.** *Suppose $\mathcal{F}_t \neq \varnothing$. Every valid center set has proposal probability $1/W_t$. Repeating the procedure and conditioning on success returns a uniform element of $\mathcal{F}_t$, including with a fixed finite attempt budget.*

*Proof.* A valid $S$ has a unique anchor satisfying $\{a\} = S \cap A_t$ and a unique distractor set $D = S \setminus \{a\}$. Its probability in one proposal is

$$\frac{w_a}{W_t} \frac{1}{\binom{|B_a|}{n-1}} = \frac{1}{W_t}.$$

Let $F_t = |\mathcal{F}_t| > 0$, $p_t = F_t/W_t$, and let $M$ be the attempt budget. The probability that a particular valid $S$ is output within $M$ attempts is

$$\sum_{j=1}^{M} (1 - p_t)^{j-1} \frac{1}{W_t} = \frac{1 - (1 - p_t)^M}{F_t}.$$

Dividing by the success probability $q_t = 1 - (1 - p_t)^M$ proves the claim. □

**Example 4.2** (Why anchors are weighted)**.** With $n = 3$, suppose two anchors have respectively two and three eligible distractors. Their proposal counts are $\binom{2}{2} = 1$ and $\binom{3}{2} = 3$. Choosing anchors with probabilities $1/4, 3/4$ makes all four proposals equally likely. Uniform anchor selection would give masses $1/2$ and $1/6$, which would bias the center sets.

The implementation permits $M = 5000$ attempts. If $W_t > 0$ but $F_t = 0$, every proposal fails; more generally, exhausting the budget does not prove that $F_t = 0$. Settings can be revised on retry while keeping the target. The uniform objects are center sets. If different centers have identical clusters, this does not imply uniformity over distinct displayed collections of clusters. Accepted cluster order and within-cluster genre order are shuffled.

**Two different notions of uniformity.** For a fixed setting, let $q_t$ be each target's probability of successful sampling, taking $q_t = 0$ when no successful proposal exists. Then

$$\mathbb{P}(T = t \mid \text{sampling succeeds}) = \frac{q_t}{\sum_{u \in V} q_u}, \qquad \sum_u q_u > 0.$$

The initial target is uniform, but targets among successfully generated games need not be. Neither uniform conditional center sets nor a retained target erases this feasibility effect. Additional conditioning on audible or completed games can create further selection effects.

### 4.1 Clues and decisions

Represent the $j$-th ordered source sample by a partial observation $a_j : V \to \mathcal{A} \cup \{\perp\}$, where $\perp$ means unavailable. The first clue is $a_1(T)$. A correct cluster choice reveals $a_2(T)$; the second decision is the exact genre within that cluster. A wrong cluster or wrong final genre ends a solo game. On revelation, the source genre page is linked.

Candidate comparison samples exclude the two clue identities, including cross-genre duplicates. Listening does not submit an answer. The geometry is constructed from coordinates and graph links, not from an embedding of the audio. The samples provide the player's evidence; the model does not assume a probability law translating sound into genre.

## 5 Genrerace: walking on the neighborhood relation

For fixed settings, define a directed movement graph

$$H_C = (V, E_C), \qquad E_C = \{(v, w) : w \in C(v),\ w \neq v\}. \tag{10}$$

A legal trace is $v_0 = s, v_1, \ldots, v_r$ with $(v_{j-1}, v_j) \in E_C$. Each accepted selection adds one vertex and one move; previewing audio adds neither. The goal is $g$, and the game ends on the first $r$ with $v_r = g$. If $s = g$, it ends with zero moves.

The shortest possible move count is

$$\delta_C(s, g) = \inf\{r : \text{a legal length-}r \text{ path from } s \text{ to } g \text{ exists}\},$$

with the infimum of the empty set taken as $+\infty$. This directed path distance is distinct from the original distance $d$. In graph mode $C(v) = B_1(v)$, so $\delta_C = d_G$. In a continuous mode, nearest-$k$ membership is generally asymmetric; $\delta_C$ can be asymmetric or infinite even though $d$ is symmetric and finite.

**Example 5.1** (A legal move into a trap, with the allowed $k = 5$)**.** Consider the seven-point subspace of the real line

$$V' = \{-1.5, -1.4, -1.3, -1.2, -1.1, 0, 1\}, \qquad d(x, y) = |x - y|.$$

For $k = 5$,

$$C_5(0) = \{0, 1, -1.1, -1.2, -1.3\}.$$

For every negative point, the five nearest points including itself are exactly the five negative points: their mutual distances are at most 0.4, whereas every distance to 0 is at least 1.1. Starting at 0 with goal 1, the move $0 \to 1$ wins immediately. But $0 \to -1.1$ is also legal and enters a closed region from which 1 cannot be reached. In particular, $0 \to -1.1$ exists while $-1.1 \to 0$ does not.

For a fixed goal define its reverse-reachable set

$$R_g = \{v \in V : \delta_C(v, g) < \infty\}.$$

A breadth-first search from $g$ on reversed movement edges computes $R_g$. The engine rejects an initial pair with $s \notin R_g$. It still displays *every* legal neighbor: a later legal move out of $R_g$ is allowed and marked as making the goal unreachable. Pruning those neighbors would change the game graph.

Each dice click samples independently from all of $V$, so the intended start/goal law is $\mathbb{P}(s = i, g = j) = N^{-2}$, with repeats allowed. The browser implements these draws using `Math.random`; the formula states the intended uniform model, not a claim of cryptographic randomness. If one conditions on the start being accepted, the pair distribution is restricted to reachable pairs. The engine does not achieve reachability by silently rerolling endpoints.

The full-space condition persists during play: $C(v_r)$ is computed relative to all $V$, not to the previously displayed cluster or the set of visited genres. Cached neighborhoods implement this same fixed graph.

# 6 Multiplayer rules

## 6.1 Genreguessr: synchronized questions and distance scores

Players in a room receive the same target and candidate clusters. All active players submit their cluster choices before the eligible players proceed to exact-genre guesses. A wrong cluster earns zero points. The result is revealed after the eligible players have submitted. The room repeats this procedure over the chosen number of questions.

For the correct cluster $C_\star$ and target $t$, define

$$D_t = \max\{d(t, u) : u \in C_\star,\ d(t, u) < \infty\}.$$

The set includes $u = t$, so this maximum is defined. It is the target's *eccentricity within the cluster*, not the pairwise diameter $\max_{u,v \in C_\star} d(u, v)$. For an eligible submitted genre $h$, the score is

$$s_t(h) = \begin{cases} 1000, & h = t \text{ or } d(t, h) = 0, \\ \operatorname{round}_{\text{even}}\left(1000 \max\{0, \min\{1, 1 - d(t, h)/D_t\}\}\right), & 0 < d(t, h) < \infty,\ D_t > 0, \\ 0, & \text{otherwise.} \end{cases} \tag{11}$$

Wrong-cluster and unsubmitted answers also receive zero. The rounding notation matches Python's nearest-integer, ties-to-even rule; stored floating-point values determine whether a numerical tie occurs. For example, $d(t, h) = D_t/2$ earns 500 points.

Scores are summed over questions. Active players with the highest total are winners, and ties are allowed. This is an explicit game reward, normalized to the particular question; it is not a universal musical similarity scale. A score of 500 in one cluster need not correspond to the same distance as 500 in another. A hypothetical duplicate XYRGB vector can earn full distance points for a different genre, although solo exact-genre success still requires the correct identifier. Rounding can also award 1000 to a sufficiently close positive-distance guess, so full points do not certify exact identification. In graph mode, $t, u \in B_1(a)$ implies $d_G(t, u) \le 2$; consequently its valid round scores are restricted to $0, 500, 1000$, unlike the finer range available with the continuous geometries.

### 6.2 Genrerace: first arrival rather than shortest route

Every player uses the same $s, g, C$, but maintains an independent legal trace. The first goal-reaching move accepted in the server's serial order ends the race and fixes the winner; subsequent moves are rejected. All final traces and move counts are then revealed. Thus the competitive objective is first arrival, not minimizing the number of moves. The shortest-path quantity $\delta_C(s, g)$ remains a benchmark rather than the implemented winner rule. When $s = g$, all starting players share a zero-move win.

## 7 Comparison and implementation boundary

| | XYRGB | Near graph | Signed kernel |
|---|---|---|---|
| Input | Map position and RGB | Near links | Near and opposite links |
| Distance | Weighted Euclidean | Edge-count shortest path | Normalized kernel chord |
| Global status | Metric if features are injective | Extended metric | Metric for $\eta > 0$ |
| Cluster | Nearest $k$, including center | Complete closed unit ball | Nearest $k$, including center |
| Race edges | Generally directed | Symmetric | Generally directed |
| Main control | Position/color balance | Source connectivity | Opposition and regularization |

These models encode different information and need not rank genre pairs in the same way. None establishes an objective ground truth of musical similarity, and the chosen cluster rule adds a second modeling decision on top of the metric.

The note's proofs concern exact arithmetic. The engine computes full $N \times N$ distances in float64, with Cholesky-based inversion for the signed kernel. It clips computed correlations to $[-1, 1]$, prevents negative squared distances from roundoff, and enforces stored symmetry. Those safeguards are not a proof that a rounded matrix satisfies every metric inequality at extreme parameter values. The sampler's uniformity proof is combinatorial: it also applies to any fixed compatibility relation obtained from the stored numerical distances.

A single dense $6291 \times 6291$ float64 matrix occupies approximately 302 MiB. Shared geometry caches avoid repeated construction for unchanged settings. Race sessions keep sparse neighborhood rows; they do not keep a separate dense matrix for each race. Cache reuse does not replace the full genre universe by a subset.

**Source and scope.** This draft was checked against repository revision `198e1ba` on 18 October 2026. The operative sources are `engine/catalog.py`, `engine/metrics.py`, `engine/sampling.py`, `engine/game.py`, `engine/race.py`, and `engine/multiplayer.py`. The repository's `DATA_SOURCES.md` records the cached crawl's provenance; `scripts/check_race_catalog.py` provides a full-catalog integration audit. The source genre map and musical samples are from Glenn McDonald's Every Noise at Once. The finite examples and propositions above are derivations for this note; they are not empirical claims about perceptual similarity.